CSE 715: Computer Graphics

Chapter 7 — Mathematics of Projection

Complete Model Answers (2020–2024)

University of Chittagong — 7^{th} Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 7 — notation follows the book exactly ($Per_{\mathbf{K}}/Par_{\mathbf{V}}$ for perspective/parallel matrices, f, θ for oblique projections, $R_0/\mathbf{N}$ for the view plane)

Contents

Theoretical Framework — Reusable Across All Years

Chapter 7 is **TIER 1** — some projection-comparison or cavalier/cabinet-matrix question appears almost every year, spread across **Q1 (short theory), Q4/Q5 (matrix derivation), Q6 (significance), and Q8 (homogeneous-coordinate derivation)** depending on the year.

Taxonomy of Projection (§7.1)

$$\text{Projections}\begin{cases}\textbf{Perspective (converging projectors)} \to \text{one-point, two-point, three-point (by num} \\ \textbf{Parallel (parallel projectors, fixed direction V)}\begin{cases}\textbf{Orthographic}(\mathbf{V} \parallel \text{view-plan} \\ \textbf{Oblique}(\mathbf{V} \nparallel \mathbf{N})\begin{cases}\text{Cavalier } (f \\ \text{Cabinet } (f\end{cases}\end{cases}\end{cases}$$

Perspective vs. Parallel: perspective projectors converge to a *center of projection* — mimics how the eye/camera sees, giving realistic depth cues but distorting true size/shape; used by **artists and photographers**. Parallel projectors stay mutually parallel — preserves true scale and shape (no foreshortening along the projection direction for orthographic types); used by **drafters, architects, and engineers** for working/technical drawings, since dimensions can be measured directly off the drawing. **Orthographic vs. Oblique** (both parallel): orthographic ⇒ direction of projection **V** is *perpendicular* to the view plane; oblique ⇒ **V** is *not* perpendicular to the view plane. **Isometric / Dimetric / Trimetric** (all orthographic axonometric — direction of projection not parallel to any principal axis, so all 3 faces show at once): **isometric** — **V** makes *equal* angles with all 3 principal axes; **dimetric** — equal angles with exactly *2* of the 3; **trimetric** — *unequal* angles with all 3. **Axonometric significance:** unlike a multiview drawing (3 separate front/top/side sheets), a single axonometric view shows all three principal faces of an object at once while still preserving parallel-projection's scale/shape fidelity along each axis — useful for technical illustration where one picture must convey full 3D form.

Perspective Projection — Math and Anomalies (§7.2)

Standard perspective transformation $Per_{\mathbf{K}}$: view plane = xy plane, center of projection $C(0,0,-d)$. By similar triangles ($\triangle ABC \sim \triangle A'OC$):

$$x' = \frac{d \cdot x}{z+d}, \qquad y' = \frac{d \cdot y}{z+d}, \qquad z' = 0$$

This mapping is *nonlinear* in 3D, but becomes linear in **homogeneous coordinates** as a 4×4 matrix:

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & d \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \qquad \text{(divide the homogeneous 4th coordinate } z{+}d \text{ back in to recover 3I}$$

Four perspective anomalies (Schaum's §7.2):

1. **Perspective foreshortening** — objects farther from the center of projection appear smaller; a large distant object and a small nearby object can project to the same apparent size.
2. **Vanishing points** — projections of lines not parallel to the view plane appear to converge at a point (e.g. railroad tracks meeting at the horizon). *Principal* vanishing points arise from lines parallel to a principal (x, y, or z) axis; the number of principal vanishing points equals the number of principal axes the view plane intersects (1, 2, or 3 ⇒ one/two/three-point perspective).
3. **View confusion** — objects located *behind* the center of projection project upside-down and backward onto the view plane.
4. **Topological distortion** — the plane through the center of projection parallel to the view plane projects to infinity; a finite line segment crossing this plane is projected as a *broken line of infinite extent*, not a normal finite segment.

Parallel Projection — Math, Oblique Form, Cavalier/Cabinet (§7.3)

Parallel projection onto the xy plane, direction $\mathbf{V} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$: since $\overrightarrow{PP'} = k\mathbf{V}$ and $z' = 0$, we get $k = -z/c$, so

$$x' = x - \frac{a}{c}z, \qquad y' = y - \frac{b}{c}z, \qquad z' = 0 \qquad \Rightarrow \qquad Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & -a/c & 0 \\ 0 & 1 & -b/c & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

General oblique projection onto the xy plane — parametrized by a foreshortening ratio f (how much a unit line perpendicular to the xy plane shrinks) and an angle θ (angle the projected perpendicular line makes with the $+x$ axis): $\mathbf{V} = f\cos\theta\,\mathbf{I} + f\sin\theta\,\mathbf{J} - \mathbf{K}$, giving

$$Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & f\cos\theta & 0 \\ 0 & 1 & f\sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Cavalier projection: $f = 1$ (no foreshortening of lines perpendicular to the xy plane). **Cabinet projection:** $f = \frac{1}{2}$ (such lines foreshortened to half length) —

cabinet therefore looks more "natural"/less elongated in depth than cavalier, at the cost of measurements along the projection direction no longer being true-length.

Examination 2024 — Q6(c,d) & Q8(a) (Section B)

Question breakdown:

Q6(c) What is the significance of Axonometric projection? (2)
Q6(d) Demonstrate some properties of Cabinet projection. (1)
Q8(a) Given a stepped/L-shaped solid block (Top/Front/Right faces labeled in the figure): draw the top, front, and right/side views. (3)

Q6(c) — Significance of Axonometric Projection [2 marks]

Axonometric projection is an **orthographic** projection (direction of projection $\perp$ view plane) chosen so that the direction of projection is *not* parallel to any of the three principal axes — unlike a multiview drawing. Its significance: **a single view reveals all three principal faces of a 3D object at once** while still preserving parallel-projection's fidelity (true scale along each axis, per the isometric/dimetric/trimetric ratio chosen) — giving a 3D impression without perspective's size-distorting foreshortening, and without needing three separate multiview sheets (front/top/side) to understand the object's shape.

Q6(d) — Properties of Cabinet Projection [1 mark]

Cabinet is an **oblique** parallel projection with foreshortening ratio $f = \frac{1}{2}$: lines perpendicular to the projection (xy) plane are drawn at **half** their true length, while lines lying in the xy plane are drawn true length/shape. The angle θ (commonly 30° or 45°) sets the direction the foreshortened depth-axis is drawn in. Because depth is foreshortened rather than left full length (as in cavalier, $f = 1$), cabinet drawings look noticeably **more realistic / less elongated** than cavalier drawings of the same object.

Q8(a) — Orthographic Multiview: Top, Front, Right/Side [3 marks]

This is a figure-only question (draw, not compute) — apply the method directly to your paper's stepped-block figure. **Method:** an orthographic multiview drawing projects the solid onto three mutually perpendicular planes using a direction of projection perpendicular to each:

1. **Front view** — look along $-z$ (or the axis labeled "Front"); draw the outline seen, with any edge hidden behind other material as a dashed line.
2. **Top view** — look straight down (along $-y$ from "Top"); draw the footprint/outline seen from above, aligned directly above the front view.
3. **Right/side view** — look along the "Right" axis; draw the outline seen from that side, aligned to the right of the front view.

For a stepped (L-shaped) block like the one shown, each view is itself a stepped/L-shaped (or rectangular, depending on which step faces that direction) outline reflecting where the block's height changes — any edge where one step is hidden behind a taller step in that viewing direction is drawn as a dashed hidden line, and all three views must line up so that corresponding features share the same width/height/depth across views (a core drafting convention).

Examination 2023 — Q1(f,i), Q4(b,c), Q5(a,c) & Q8(b)

Question breakdown:

- Q1(f) You intend to draw a picture of your university — what kind of projection technique will you consider? (≈1, part of 9×1)
- Q1(i) Differentiate among the isometric, dimetric, and trimetric projections. (≈1)
- Q4(b) Differentiate the orthographic and oblique projection. (1)
- Q4(c) Explain the four anomalies of perspective projection with necessary diagrams. (4)
- Q5(a) Find the transformation for (a) cavalier with $\theta = 60°$ and (b) cabinet projections with $\theta = 45°$. (3.5)
- Q5(c) Derive the equation of parallel projection onto the xy plane in the direction of projection $\mathbf{V} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$. (3)
- Q8(b) Derive the matrix for performing a perspective projection using homogeneous coordinates. (2)

Q1(f) — Projection Choice for a Picture of the University

A "picture" (photographic/realistic depiction) of a real building is best captured with **perspective projection** — it matches how the human eye/a camera actually sees the scene, including natural depth cues (foreshortening, converging vanishing lines on receding building edges). Parallel/orthographic projection would instead be the right choice only if the goal were an accurate-to-scale technical/architectural drawing rather than a realistic picture.

Q1(i) — Isometric vs. Dimetric vs. Trimetric

See §1.1: **isometric** = equal angles with all 3 principal axes; **dimetric** = equal angles with exactly 2 of the 3 (third differs); **trimetric** = all 3 angles unequal. All three are orthographic *axonometric* projections (direction of projection not parallel to any single principal axis).

Q4(b) — Orthographic vs. Oblique Projection [1 mark]

Both are parallel (fixed-direction-projector) projections. **Orthographic:** direction of projection $\mathbf{V}$ is *perpendicular* to the view plane ($\mathbf{V} \parallel \mathbf{N}$) — includes multiview and axonometric (isometric/dimetric/trimetric). **Oblique:** direction of projection is *not* perpendicular to the view plane — includes cavalier ($f = 1$) and cabinet ($f = \frac{1}{2}$).

Q4(c) — Four Anomalies of Perspective Projection [4 marks]

See §1.2 for the full list with diagrams: (1) perspective foreshortening (farther = smaller), (2) vanishing points (non-view-plane-parallel lines appear to converge), (3) view confusion (objects behind the center of projection appear upside-down/backward), (4) topological distortion (a finite segment crossing the plane-through-the-center-of-projection-parallel-to-the-view-plane projects to an infinite broken line).

Q5(a) — Cavalier $\theta = 60^\circ$, Cabinet $\theta = 45^\circ$ [3.5 marks]

Using $Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & f\cos\theta & 0 \\ 0 & 1 & f\sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$:

Cavalier ($f = 1,\ \theta = 60^\circ$): $\cos 60^\circ = \frac{1}{2},\ \sin 60^\circ = \frac{\sqrt{3}}{2}$.

$$Par_{\text{cav}} = \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Cabinet ($f = \frac{1}{2},\ \theta = 45^\circ$): $\cos 45^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2}$, so $f\cos\theta = f\sin\theta = \frac{\sqrt{2}}{4}$.

$$Par_{\text{cab}} = \begin{pmatrix} 1 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 1 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Q5(c) — Derive Parallel Projection, $\mathbf{V} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$ [3 marks]

From Fig. 7-22: $\overrightarrow{PP'}$ has the same direction as $\mathbf{V}$, so $\overrightarrow{PP'} = k\mathbf{V}$ for some scalar k. Comparing components: $x' - x = ka$, $y' - y = kb$, $z' - z = kc$. Since the image lies on the xy plane, $z' = 0 \Rightarrow k = -z/c$. Substituting:

$$x' = x - \frac{a}{c}z, \qquad y' = y - \frac{b}{c}z, \qquad z' = 0$$

$$Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & -a/c & 0 \\ 0 & 1 & -b/c & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad P' = Par_{\mathbf{V}} \cdot P$$

Q8(b) — Perspective Projection Matrix via Homogeneous Coordinates [2 marks]

Standard perspective transformation: view plane = xy plane, center of projection $C(0, 0, -d)$. By similar triangles $\triangle ABC \sim \triangle A'OC$ (Fig. 7-4): $x' = \dfrac{d \cdot x}{z + d}$, $y' = \dfrac{d \cdot y}{z + d}$, $z' = 0$. This is nonlinear in ordinary 3D coordinates, but linear in homogeneous coordinates $(x, y, z, 1)$:

$$\begin{pmatrix} x'_h \\ y'_h \\ z'_h \\ h \end{pmatrix} = \begin{pmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & d \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} dx \\ dy \\ 0 \\ z + d \end{pmatrix}$$

Dividing through by the homogeneous coordinate $h = z + d$ recovers $x' = \dfrac{dx}{z + d}$, $y' = \dfrac{dy}{z + d}$, $z' = 0$ — confirming the matrix is correct. This is exactly how a nonlinear perspective divide is made representable as ordinary matrix multiplication.

Examination 2022 — Q5(a,b) (Section B)

Question breakdown:

Q5(a) Differentiate between perspective and parallel projection. Which projection is used by architects and engineers? (1)

Q5(b) The unit cube is projected onto the XY plane. Find the transformation and projection matrix for (a) cavalier with $\theta = 30°$ and (b) cabinet projections with $\theta = 45°$. (4)

Q5(a) — Perspective vs. Parallel, Architects/Engineers [1 mark]

See §1.1: perspective converges to a center of projection (realistic, used by artists/photographers, distorts true size via foreshortening); parallel keeps projectors mutually parallel (preserves true scale/shape, used by **architects and engineers** for working drawings where measurements must be read directly off the page).

Q5(b) — Cavalier $\theta = 30^\circ$, Cabinet $\theta = 45^\circ$ [4 marks]

Cavalier ($f = 1,\ \theta = 30^\circ$): $\cos 30^\circ = \frac{\sqrt{3}}{2},\ \sin 30^\circ = \frac{1}{2}$.

$$Par_{\text{cav}} = \begin{pmatrix} 1 & 0 & \frac{\sqrt{3}}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Cabinet ($f = \frac{1}{2},\ \theta = 45^\circ$): identical numeric result to 2023's cabinet sub-part above (same angle):

$$Par_{\text{cab}} = \begin{pmatrix} 1 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 1 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Applying either matrix to the unit-cube homogeneous vertex matrix $V = (ABCDEFGH)$ (Schaum's Fig. 7-12) draws the projected cube by adding the $f\cos\theta, f\sin\theta$ offset to the back-face vertices E, F, G, H while the front face A, B, C, D is unchanged (since it already lies in the xy plane).

Examination 2021 — Q5(a,b,c) (Section B)

Question breakdown:

- Q5(a) Distinguish between perspective and parallel projection. Which projection is used by architects and engineers? (1.5)
- Q5(b) The unit cube is projected onto the XY plane. Find the transformation and projection matrix for (a) cavalier with $\theta = 45^\circ$ and (b) cabinet projections with $\theta = 30^\circ$. (4)
- Q5(c) What do you mean by perspective foreshortening and vanishing point? Explain with the necessary diagram. (2)

Q5(a) — Perspective vs. Parallel, Architects/Engineers [1.5 marks]

Identical answer to 2022 Q5(a) and 2020 Q8(a) above — see §1.1.

Q5(b) — Cavalier $\theta = 45°$, Cabinet $\theta = 30°$ [4 marks]

Cavalier ($f = 1,\ \theta = 45°$): $\cos 45° = \sin 45° = \frac{\sqrt{2}}{2}$.

$$Par_{\text{cav}} = \begin{pmatrix} 1 & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 1 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Cabinet ($f = \frac{1}{2},\ \theta = 30°$): $\cos 30° = \frac{\sqrt{3}}{2},\ \sin 30° = \frac{1}{2} \Rightarrow f\cos\theta = \frac{\sqrt{3}}{4},\ f\sin\theta = \frac{1}{4}$.

$$Par_{\text{cab}} = \begin{pmatrix} 1 & 0 & \frac{\sqrt{3}}{4} & 0 \\ 0 & 1 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Q5(c) — Perspective Foreshortening and Vanishing Point [2 marks]

Perspective foreshortening: the illusion that objects/lengths appear smaller the farther they are from the center of projection (Fig. 7-5 — a sphere $2\frac{1}{2}\times$ larger but twice as far away appears the *same size* as a nearer, smaller sphere). **Vanishing point:** the illusion that a family of parallel lines *not* parallel to the view plane appears, under perspective projection, to converge to a single point on the view plane (e.g. railroad tracks appearing to meet at the horizon) — formally, the projection of the point at infinity in the lines' common direction.

Examination 2020 — Q6(d) & Q8(a,b) (Group-B)

Question breakdown:

Q6(d) What do you mean by perspective foreshortening and vanishing point? (1.75)
Q8(a) Distinguish between perspective and parallel projections. Which projection is used by the architects and engineers? (3)
Q8(b) Draw the diagrams of isometric, diametric, and trimetric projections. (2.25)

Q6(d) — Perspective Foreshortening and Vanishing Point [1.75 marks]

Identical answer to 2021 Q5(c) above — see §1.2 and the 2021 Q5(c) solution.

Q8(a) — Perspective vs. Parallel, Architects/Engineers [3 marks]

Identical answer to 2021/2022 Q5(a) above — see §1.1. (Highest mark-weight instance of this recurring question — write the full comparison: converging vs. parallel projectors, realistic depth vs. true-scale preservation, artists/photographers vs. architects/engineers.)

Q8(b) — Isometric / Diametric / Trimetric Diagrams [2.25 marks]

Draw three cubes, each showing all 3 faces (axonometric — direction of projection not parallel to any axis): **isometric** — all 3 visible edge-angles from the corner are equal (120° apart in the classic top-corner view); **dimetric** — exactly 2 of the 3 angles/axis-foreshortenings are equal, the third distinct; **trimetric** — all 3 angles/foreshortenings are different. See §1.1 for the formal angle definitions.

Quick Summary — Exam Patterns for Chapter 7

Pattern	Years	Method to memorise
Perspective vs. Parallel — distinguish + architects/engineers (exact classic phrasing)	2020, 2021, 2022	Converging vs. parallel projectors; realistic-but-distorted vs. true-scale; artists/photographers vs. architects/engineers
Cavalier/Cabinet projection matrix for given θ	2021, 2022, 2023	$Par_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & f\cos\theta & 0 \\ 0 & 1 & f\sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, cavalier $f = 1$, cabinet $f = \frac{1}{2}$ — just plug in θ
Perspective foreshortening + vanishing point (explain)	2020, 2021	Foreshortening = farther looks smaller; vanishing point = non-view-plane-parallel lines converge to a point
Isometric/Dimetric/Trimetric — differentiate or draw	2020, 2023	Equal angles with all 3 axes / exactly 2 / none — isometric/dimetric/trimetric respectively
Orthographic vs. Oblique projection	2023 only	Direction of projection $\perp$ view plane (ortho) vs. not (oblique)
Perspective anomalies (4, with diagrams)	2023 only	Foreshortening, vanishing points, view confusion, topological distortion
General parallel/perspective projection derivation (homogeneous coordinates)	2023 only	$Par_{\mathbf{V}}$: $x' = x - \frac{a}{c}z$ etc.; $Per_{\mathbf{K}}$: $x' = \frac{dx}{z+d}$ made linear via the homogeneous 4th row
Axonometric significance / Cabinet properties (light-touch)	2024 only	One view shows all 3 faces; cabinet halves perpendicular-line length vs. cavalier's no foreshortening
Orthographic multiview (top/front/side) drawing	2024 only	Project along each of 3 perpendicular directions; hidden edges dashed; views aligned

Exam-day strategy for Chapter 7: correction from earlier tracking — "Perspective vs. Parallel projection" and "Cavalier/Cabinet" were both previously logged as 5/5-year topics; re-verified directly against all pages of all 5 raw scanned papers on 2026-07-05. Cavalier/Cabinet's exact matrix-derivation question is genuinely only **2021, 2022, 2023** (2020 has no such question at all; 2024 only asks for "properties of Cabinet" qualitatively, no θ given). The classic "distinguish perspective vs. parallel + architects/engineers" phrasing is **2020, 2021, 2022** only — 2023/2024 ask related-but-differently-shaped projection questions instead (orthographic/oblique, projection choice for a scenario, axonometric significance). The underlying *concept cluster* (some projection-type comparison) is still 5/5 — just don't expect the identical wording every year.